

FINITE ELEMENT ANALYSIS OF A 2D METAL PLATE USING MATLAB

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Abstract

This paper investigates the accuracy of Finite Element Analysis using commercial engineering applications, Matlab and ANSYS, to compare result. Programming language for a two-dimensional linear elastic metal plate made of Aluminium alloy consists of two and four; 3-noded and 6-noded triangular elements under uniform tension was used for analysing displacement and strain. Three additional Young's modulus were examined, 148 GPa, 200 GPa, and 300 GPa. Also, three different load conditions were applied, $\theta = 30^\circ$, $\theta = 60^\circ$, and $\theta = 110^\circ$. The aim of this study is to examine the accuracy of displacement and strain obtained from the FEM from the code developed and validate it with ANSYS result.

1. Introduction

Structures such as bridges and beams generally consist of number of elements which is assembled together. Typically, bridges and beams are supported at one end and free at the other end to allow material deformation resulting from thermal expansion and high loading. Therefore, it is essential to carry out stress analysis to define the critical points of the structure design and restrict it with boundary conditions to avoid failure. Also, similar test is required for the selected material to study the behaviour of it which provides the maximum allowable stress that can be applied on the material and hence its maximum deformation. These tests can be carried out experimentally but its required an enormous amount of effort and money particularly for complicated designs. However, due to the development of technology, FEM can be used through different commercial engineering applications such as ANSYS, which allows the user to analyse all the points of interest on the model with less effort and much cheaper comparing to experimental test. FEM was established to study the stresses in complex structures by solving large number of simultaneous equations.

In this report, Matlab R2016 and ANSYS v15 mechanical were used as a tool to compute the Finite Element Approach to analyse displacement and strain of a rectangular metal plate under uniform tensile stress with a uniform thickness. The plate is simply supported at two nodes as shown in figure 1 and consists of two and four triangular elements with 3 and 6 nodes. The purpose of this study is to compare the displacement and strain of the plate for various elements, material properties, and load conditions. The accuracy of the code will be validated using ANSYS result. Model description, material properties, boundary conditions, and element type/shape will be illustrated in the report as well as the equations used. In addition, results obtained from the code and ANSYS will be evaluated and relationship between strain and elastic modulus will be achieved.

2. Methodology:

a. Description of the model

The model used consist of a uniform rectangular plate that has a height of 0.03 m and a width of 0.05 m with a thickness of 0.002 m.

b. Material selection

Isotropic Aluminium alloy with elastic modulus of 100 GPa and Poisson's ratio of 0.3 was initially implemented. In addition, different elastic materials have been tested in this study such as Copper alloy with $E = 148$ GPa, Low Carbon Steel with $E = 200$ GPa, and Silicon Carbide with $E = 300$ GPa.

c. Boundary conditions and load

The plate is fixed supported ($UX=UY=0$) at the bottom point on the left-hand side and roller support was used for the top point, which permit the movement in the y-direction ($UX=0$). On the other hand, tensile stress of 200 MPa is applied uniformly in the x-direction at 90° as represented in figure 1. The direction of load has been altered three different times from 90° to $\theta = 30^\circ$, $\theta = 60^\circ$, and $\theta = 110^\circ$ with keeping other variables constant.

d. Element type

3-noded and 6-noded triangular elements will be used in this study. In order to model the plate in ANSYS, 2-D structural solid mass quad 4-node PLANE 182 with triangular element shape was used for the 3-noded element method. This element type can be used as either plane stress/strain or axisymmetric element. Each node has two DOF, x and y. It has plasticity, large deflection, and strain capability. PLANE 183 quad 8 was used for modelling the 6-noded elements because of its high order nodes. This element type has quadratic displacement behaviour and typically used for modelling irregular meshes. Like PLANE 182, quad 8 has two DOF at each node. Figure 1 part 2 shows the 3 and 6 noded elements order as it has been done in the code and figure 2 represents the boundary conditions and load applied in ANSYS.

e. Coding and simulation

Programming language requires from the user to define all the model variables such as dimensions, loads, and material properties. After this the FEM can be computed using the same language for both 3-node and 6-node elements. The level of complexity of the code depends on the designer skills. It is recommended to use Matlab for this study. Also, the same approach has been applied on ANSYS to examine the validity of the code.

f. Equations and measurements

- 3-noded triangular element:

The plate coordinates are plotted as shown in figure 1. In order to calculate the strain and displacement of the element it is important to obtain the stiffness matrix K for each element. This can be calculated by the following equation:

$$[k] = tA[B]^T[D][B] \quad (1)$$

where t is the thickness of the model and A is the area of each element which can be defined as:

$$A = \frac{1}{2} * base * height \quad (2)$$

and the strain-displacement matrix B is constructed as follows:

$$[B] = \frac{1}{2A} \begin{bmatrix} \beta_1 & 0 & \beta_2 & 0 & \beta_3 & 0 \\ 0 & \gamma_1 & 0 & \gamma_2 & 0 & \gamma_3 \\ \gamma_1 & \beta_1 & \gamma_2 & \beta_2 & \gamma_3 & \beta_3 \end{bmatrix} \quad (3)$$

where β and γ are the shape function coefficients, and can be calculated as shown in equation 4.

$$\begin{aligned} \beta_1 &= y_2 - y_3 & \gamma_1 &= x_2 - x_3 \\ \beta_2 &= y_3 - y_1 & \gamma_2 &= x_3 - x_1 \\ \beta_3 &= y_1 - y_2 & \gamma_3 &= x_2 - x_1 \end{aligned} \quad (4)$$

the stress-strain matrix D :

$$[D] = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1 - 2\nu}{2} \end{bmatrix} \quad (5)$$

where ν is the Poisson's ratio of the material. Therefore, the displacement was obtained from:

$$[F] = [k][U] \quad (6)$$

the $[K]$ matrix has been reduced due to the boundary conditions using Gaussian elimination and F is the force matrix which contains the total force applied on the plate and can be calculated from this relationship:

$$\sigma = F/A$$

where ($A = \text{height} \times \text{thickness}$) the area of the plate in the z direction. After obtaining the total force applied, it will be divided by the number of nodes in which the load is acting. Hence, the strain can be calculated as follows:

$$[\varepsilon] = [B][U] \quad (7)$$

- 6-noded triangular element:

The stiffness matrix of the quadratic triangular element requires the calculation of three $[B]$ and $[K]$ matrices. The method used to calculate the displacement and strain in the 6-noded element will be as follows:

$$[K]^e = \int_V [B]^T [D] [B] dv = \int_0^1 \int_{-0}^{1-L1} [B]^T [D] [B] t |J| dL1 dL2 \quad (8)$$

or $[K]^e = H \cdot t \cdot [B]^T [D] [B] \det[J]$, where H is the weighting factor = 1/6

where [D] can be calculated from equation 5, and the strain-displacement matrix $[B]_{3 \times 18}$ is calculated from:

$$[B]_{3 \times 18} = [B_1]_{3 \times 4} [B_2]_{4 \times 4} [B_3]_{4 \times 18} \quad (9)$$

where $[B_1]$, $[B_2]$, and $[B_3]$ are obtained using the derivative shape functions matrix and Jacobian matrix:

$$[B_1] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

$$[B_2] = \begin{bmatrix} \frac{J_{22}}{|J|} & -\frac{J_{12}}{|J|} & 0 & 0 \\ -\frac{J_{21}}{|J|} & \frac{J_{11}}{|J|} & 0 & 0 \\ 0 & 0 & \frac{J_{22}}{|J|} & -\frac{J_{12}}{|J|} \\ 0 & 0 & -\frac{J_{21}}{|J|} & \frac{J_{11}}{|J|} \end{bmatrix}$$

$$[B_3] = \begin{bmatrix} \frac{\partial N_1}{\partial L_1} & 0 & \frac{\partial N_2}{\partial L_1} & 0 & \frac{\partial N_3}{\partial L_1} & 0 & \frac{\partial N_4}{\partial L_1} & 0 & \frac{\partial N_5}{\partial L_1} & 0 & \frac{\partial N_6}{\partial L_1} & 0 \\ \frac{\partial N_1}{\partial L_2} & 0 & \frac{\partial N_2}{\partial L_2} & 0 & \frac{\partial N_3}{\partial L_2} & 0 & \frac{\partial N_4}{\partial L_2} & 0 & \frac{\partial N_5}{\partial L_2} & 0 & \frac{\partial N_6}{\partial L_2} & 0 \\ 0 & \frac{\partial N_1}{\partial L_1} & 0 & \frac{\partial N_2}{\partial L_1} & 0 & \frac{\partial N_3}{\partial L_1} & 0 & \frac{\partial N_4}{\partial L_1} & 0 & \frac{\partial N_5}{\partial L_1} & 0 & \frac{\partial N_6}{\partial L_1} \\ 0 & \frac{\partial N_1}{\partial L_2} & 0 & \frac{\partial N_2}{\partial L_2} & 0 & \frac{\partial N_3}{\partial L_2} & 0 & \frac{\partial N_4}{\partial L_2} & 0 & \frac{\partial N_5}{\partial L_2} & 0 & \frac{\partial N_6}{\partial L_2} \end{bmatrix}$$

And the quadratic shape functions in natural coordinates are given by:

$$N_1 = L_1(2L_1 - 1)$$

$$N_2 = L_2(2L_2 - 1)$$

$$N_3 = (1 - L_1 - L_2)(1 - 2L_1 - 2L_2)$$

$$N_4 = 4L_1L_2$$

$$N_5 = 4L_2(1 - L_1 - L_2)$$

$$N_6 = 4L_1(1 - L_1 - L_2)$$

derivatives of shape functions are given by:

$$\begin{aligned}
 \frac{\partial N_1}{\partial L_1} &= 4L_1 - 1 & \frac{\partial N_1}{\partial L_2} &= 0 \\
 \frac{\partial N_2}{\partial L_1} &= 0 & \frac{\partial N_2}{\partial L_2} &= 4L_2 - 1 \\
 \frac{\partial N_3}{\partial L_1} &= -3 + 4L_1 + 4L_2 & \frac{\partial N_3}{\partial L_2} &= -3 + 4L_1 + 4L_2 \\
 \frac{\partial N_4}{\partial L_1} &= 4L_2 & \frac{\partial N_4}{\partial L_2} &= 4L_1 \\
 \frac{\partial N_5}{\partial L_1} &= -4L_2 & \frac{\partial N_5}{\partial L_2} &= 4 - 4L_1 - 8L_2 \\
 \frac{\partial N_6}{\partial L_1} &= 4 - 4L_2 - 8L_1 & \frac{\partial N_6}{\partial L_2} &= -4L_1
 \end{aligned}$$

The Jacobian for the isoparametric element is:

$$[J] = \begin{bmatrix} \sum_{i=1}^6 \frac{\partial N_i}{\partial L_1} x_i & \sum_{i=1}^6 \frac{\partial N_i}{\partial L_1} y_i \\ \sum_{i=1}^6 \frac{\partial N_i}{\partial L_2} x_i & \sum_{i=1}^6 \frac{\partial N_i}{\partial L_2} y_i \end{bmatrix}$$

Therefore, the displacement and strain can be calculated from:

$$\{\varepsilon\}_{3 \times 1} = [B]_{3 \times 18} \{\delta\}_{18 \times 1}^e \quad (10)$$

And the displacement is found from the relationship in equation 6. However, the forces will be distributed on three nodes and will be, 1/6 the total force for the top and bottom nodes and 2/3 for the middle node.

g. Validity of the program

The code was validated by comparing the displacement and strain obtained with ANSYS result. The plate was modelled in ANSYS and the boundary conditions and load from the program were applied in a similar manner on ANSYS. The result was fairly satisfying; however, the order of the nodes is different in ANSYS than the program as well as the decimal places.

3. Results

The stiffness matrices obtained from the program for one element for both, 3-noded and 6-noded triangular, are shown in figure 3.

The code has been extended to solve the displacement and strain of two triangular elements. Table 1 represents the comparison of displacement and strain obtained from the code with ANSYS result for the two and four 3-noded triangular elements. In addition, table 2 compares the displacement and strain obtained from the code with ANSYS for the two and four 6-noded triangular elements. The deformation shape of the plate for both cases are shown in Figure 4.

Moreover, the code has been tested for different materials such as Copper alloy ($E=148$ GPa), Low Carbon steel ($E=200$ GPa), and Silicon Carbide ($E=300$ GPa). The calculated displacement and strain of these materials based on the four 6-noded triangular elements are summarised in figure 5. Furthermore, displacement and strain have been computed when the angle of the load applied is altered i.e. $\theta = 30^\circ$, $\theta = 60^\circ$, and $\theta = 110^\circ$, the effect of orientated load is shown in figure 5.

4. Discussion

Figure 3 part (a) represents the stiffness matrix for a 3-noded triangular element and part (b) for the 6-noded element. These matrices were expanded to the global coordinate in order to find the displacements, forces and strains of the element.

The result listed in table 1 in accordance to the global coordinate arrangement in figure 1 and 2, proves the validity of the code for the two and four 3-noded elements. However, ANSYS result were more accurate than the code, for instance, ANSYS provides more decimals in the solution while the code gives zeros. It can be noticed that the 4 elements solution has more nodes, hence additional degree of freedom which gives more accurate result than the 2 elements problem. In addition, the two and four 6-noded elements result for displacement and strain are summarised in table 2. The values of displacement obtained from the code are identical to ANSYS, which validates the accuracy of the code. Nevertheless, the strain values have a high percentage of error, average of 110%, which dictates that the code has some limitations when solving quadratic (6-noded) problems. Also, it is hard to find analytical form of integrals, equation 8, and the jacobian calculation depends on the selection of shape functions, which depends on the choice of elements. If the values of nodes added up for each element, it will give a fairly close value comparing to ANSYS.

Furthermore, the four 6-noded triangular elements code was tested for 3 additional elastic moduli as mentioned in the result section. The relationship between the elastic moduli of the material and the strain is found to be inverse, low Young's modulus has small strain and small displacement, as shown in figure 5 (a). While the relationship between stress and strain is proportional as shown in figure 5 (c). Moreover, the angle of the load applied was altered three different times, and the effect of it was recognised and plotted in figure 5 (b). The maximum force can be achieved on the X-direction when the load is applied at $\theta = 90^\circ$, and hence maximum strain. While the minimum can be obtained at $\theta = 0^\circ$ and $\theta = 180^\circ$. On the other hand, the maximum strain and displacement can be achieved at $\theta \approx 0^\circ$ to 15° for the Y-direction forces and the minimum is at $\theta = 180^\circ$.

Generally, the four 6-noded triangular elements have more accurate result than the 3-noded (linear) problem due to the high number of simultaneous equations which allows the model to displace in a flexible manner as shown in figure 4 (c) and (d).

5. Conclusion

The two and four 3-noded elements (linear) problem has less accurate result than the 6-noded one. The code has some limitations when solving high order quadratic problems as shown in table 2 for the strain. The strain is directly affected by the angle of the applied load as well as the elastic moduli of the material used. High order elements provide accurate result and allow flexibility when measuring the displacement or strain in a particular model.

References

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Figures and tables

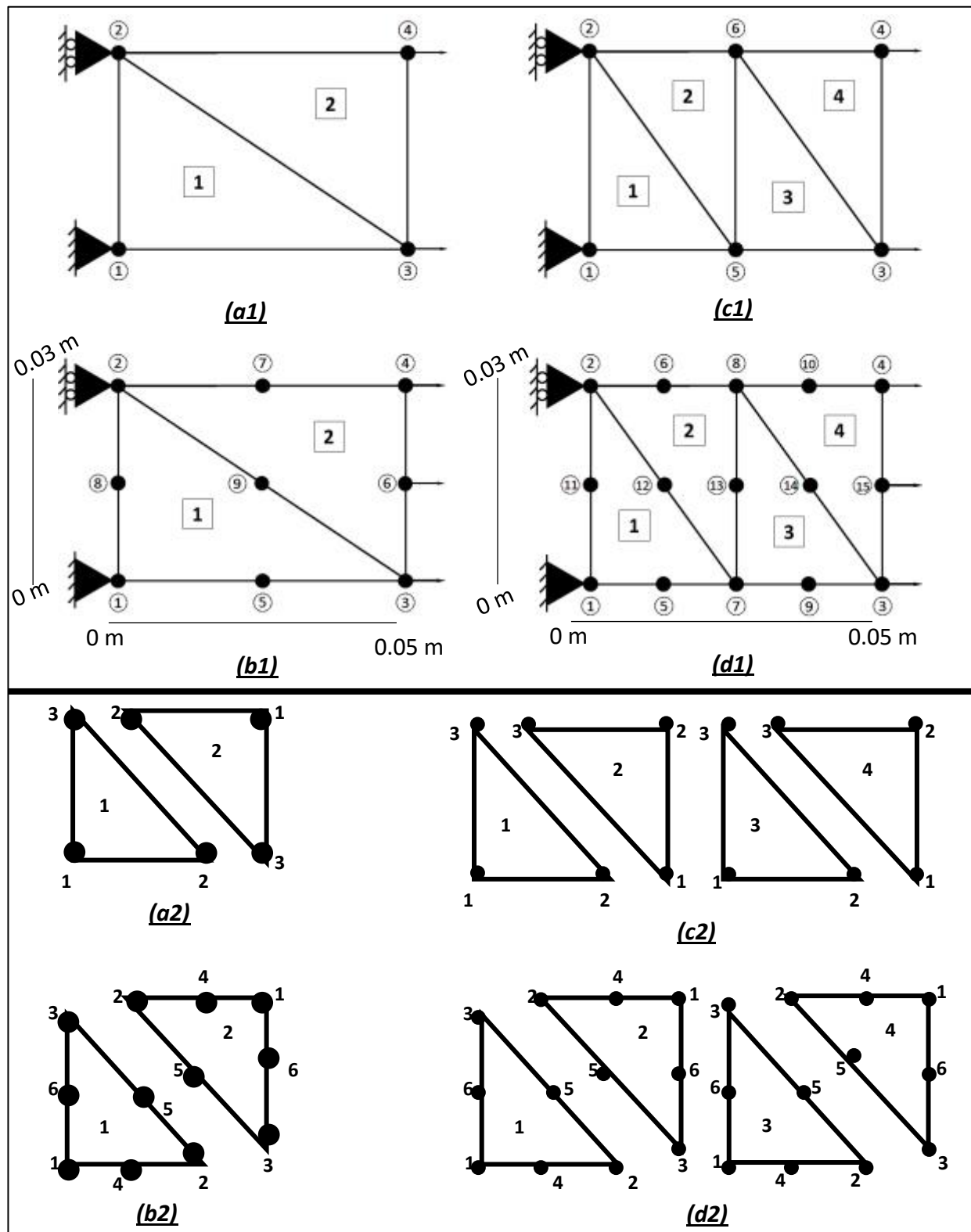


Figure 1. Part 1 - (a1) shows the global nodes ordering for the 3-noded 2 elements triangular and (c1) is the 4 elements. (b1) represents the global nodes of the 6-noded 2 elements triangular and (d1) is the 4 elements. Part 2 - (a2) shows the local nodes arrangements of the 3-noded 2 elements and (c2) represents the 4 elements. (b2) and (d2) shows the 6-noded 2 and 4 elements local nodes arrangements.

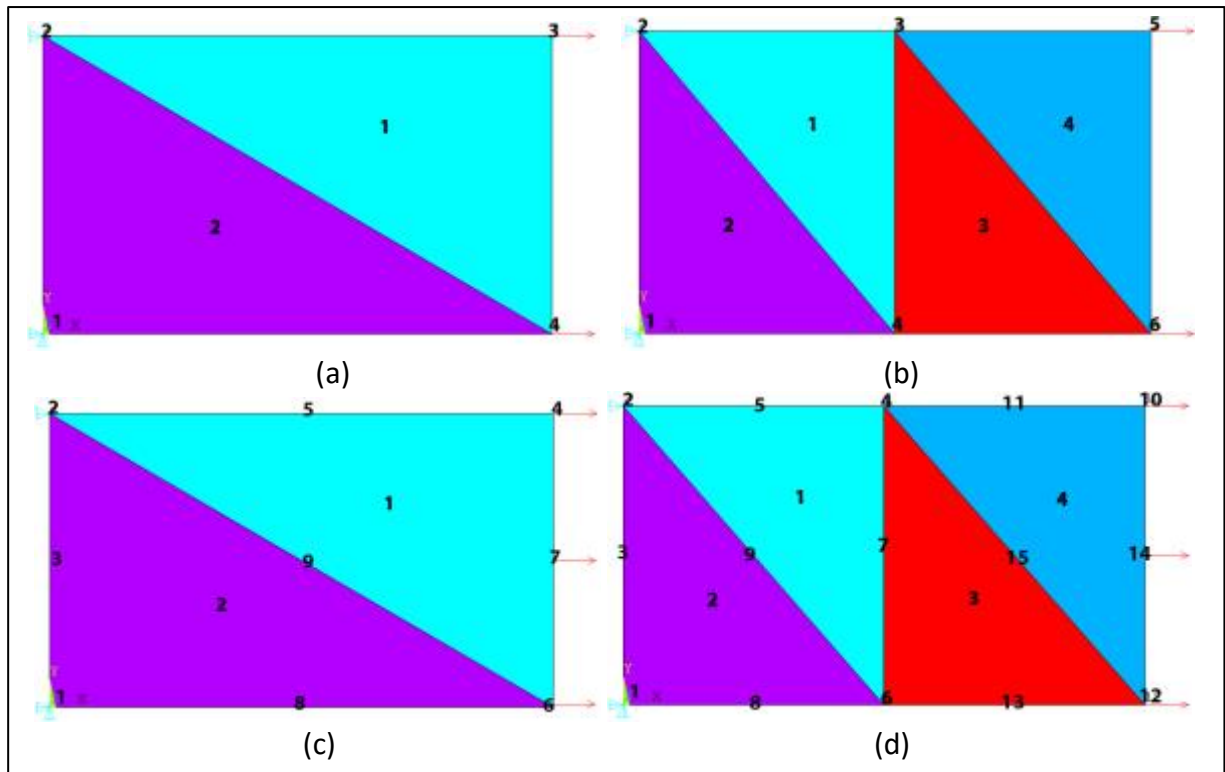


Figure 2. the boundary conditions and loads applied using ANSYS. (a) and (b) show the 3-noded 2 and 4 elements problem. While (c) and (d) represent the 6-noded 2 and 4 elements problem. The number of nodes represent the global coordinate for ANSYS solution.

k1 =						(a)					
1.0e+08 *											
1.3004	0.7143	-0.6593	-0.3846	-0.6410	-0.3297						
0.7143	2.0623	-0.3297	-0.2308	-0.3846	-1.8315						
-0.6593	-0.3297	0.6593	0	0	0.3297						
-0.3846	-0.2308	0	0.2308	0.3846	0						
-0.6410	-0.3846	0	0.3846	0.6410	0						
-0.3297	-1.8315	0.3297	0	0	1.8315						

K =												(b)									
1.0e+08 *																					
1.3004	0.7143	0.2198	0.1282	0.2137	0.1099	-0.8791	-0.5128	0.0000	0	-0.8547	-0.4396	1.3004	0.7143	0.2198	0.1282	0.2137	0.1099	-0.8791	-0.5128	0.0000	0
0.7143	2.0623	0.1099	0.0769	0.1282	0.6105	-0.4396	-0.3077	0	0	-0.5128	-2.4420	0.7143	2.0623	0.1099	0.0769	0.1282	0.6105	-0.4396	-0.3077	0	0
0.2198	0.1099	0.6593	0	-0.1099	-0.8791	-0.4396	0	0.4396	0	0	0	0.2198	0.1099	0.6593	0	-0.1099	-0.8791	-0.4396	0	0.4396	0
0.1282	0.0769	0	0.2308	-0.1282	0	-0.5128	-0.3077	0.5128	0	0	0	0.1282	0.0769	0	0.2308	-0.1282	0	-0.5128	-0.3077	0.5128	0
0.2137	0.1282	0	-0.1282	0.6410	0	0	0	0	0.5128	-0.8547	-0.5128	0.2137	0.1282	0	-0.1282	0.6410	0	0	0	0	-0.5128
0.1099	0.6105	-0.1099	0	0	1.8315	0	0	0.4396	0	-0.4396	-2.4420	0.1099	0.6105	-0.1099	0	0	1.8315	0	0	0.4396	0
-0.8791	-0.4396	-0.8791	-0.5128	0	0	3.4676	0.9524	-1.7094	-0.9524	0.0000	0.9524	-0.8791	-0.4396	-0.8791	-0.5128	0	0	3.4676	0.9524	-1.7094	-0.9524
-0.5128	-0.3077	-0.4396	-0.3077	0	0	0.9524	5.4994	-0.9524	-4.8840	0.9524	0	-0.5128	-0.3077	-0.4396	-0.3077	0	0	0.9524	5.4994	-0.9524	-4.8840
0.0000	0	0	0.5128	0	0.4396	-1.7094	-0.9524	3.4676	0.9524	-1.7582	-0.9524	0.0000	0	0	0.5128	0	0.4396	-1.7094	-0.9524	3.4676	0.9524
0	0	0.4396	0	0.5128	0	-0.9524	-4.8840	0.9524	5.4994	-0.9524	-0.6154	0	0	0.4396	0	0.5128	0	-0.9524	-4.8840	0.9524	-0.6154
-0.8547	-0.5128	0	0	-0.8547	-0.4396	0.0000	0.9524	-1.7582	-0.9524	3.4676	0.9524	-0.8547	-0.5128	0	0	-0.8547	-0.4396	0.0000	0.9524	-1.7582	-0.9524
-0.4396	-2.4420	0	0	-0.5128	-2.4420	0.9524	0	-0.9524	-0.6154	0.9524	5.4994	-0.4396	-2.4420	0	0	-0.5128	-2.4420	0.9524	0	-0.9524	-0.6154

Figure 3. (a) represents the stiffness matrix obtained for the 3-noded element and (b) is the [K] for the 6-noded element model.

3-noded Displacement						3-noded Strain					
		2 elements		4 elements				2 elements		4 elements	
Node		Matlab	ANSYS	Matlab	ANSYS	Element		Matlab	ANSYS	Matlab	ANSYS
1	u	0	0	0	0	1	ϵ_x	0.002	2.00E-03	0.002	2.00E-03
	v	0	0	0	0		ϵ_y	-0.0006	-6.00E-04	-0.0006	-6.00E-04
2	u	0	0	0	0		γ_{xy}	0.0000	5.96E-19	-0.0000	-1.35E-18
	v	-1.80E-05	-1.80E-05	-1.80E-05	-1.80E-05	2	ϵ_x	0.002	2.00E-03	0.002	2.00E-03
3	u	1.00E-04	1.00E-04	1.00E-04	5.00E-05		ϵ_y	-0.0006	-6.00E-04	-0.0006	-6.00E-04
	v	-0.0000	-1.80E-05	-0.0000	-1.80E-05		γ_{xy}	-0.0000	-2.10E-19	-0.0000	-2.10E-19
4	u	1.00E-04	1.00E-04	1.00E-04	5.00E-05	3	ϵ_x			0.002	2.00E-03
	v	-1.80E-05	5.08E-20	-1.80E-05	-4.27E-20		ϵ_y			-0.0006	-6.00E-04
5	u			5.00E-05	1.00E-04		γ_{xy}			-0.0000	-1.31E-18
	v			-0.0000	-1.80E-05	4	ϵ_x			0.002	2.00E-03
6	u			5.00E-05	1.00E-04		ϵ_y			-0.0006	-6.00E-04
	v			-1.80E-05	-8.81E-20		γ_{xy}			-0.0000	-1.00E-18

Table 1. comparison between Matlab result and ANSYS for displacement and strain obtained from the 3-noded; 2 and 4 elements.

6-noded Displacement						6-noded Strain					
		2 elements		4 elements				2 elements		4 elements	
Node		Matlab	ANSYS	Matlab	ANSYS	Element		Matlab	ANSYS	Matlab	ANSYS
1	u	0	0	0	0	1	ϵ_x	0.005	3.37E-03	0.0055	5.11E-03
	v	0	0	0	0		ϵ_y	-0.0018	-1.98E-03	-0.0027	-3.59E-03
2	u	0	0	0	0		γ_{xy}	0.0008	4.60E-03	0.0025	4.90E-03
	v	-3.43E-05	-3.43E-05	-5.81E-05	-5.81E-05	2	ϵ_x	0.0065	2.02E-03	0.0067	2.23E-03
3	u	1.26E-04	4.01E-05	1.40E-04	6.07E-05		ϵ_y	-0.0018	-5.99E-04	-0.0016	-6.20E-04
	v	-1.11E-05	-2.40E-05	-2.52E-05	-4.14E-05		γ_{xy}	-0.0008	-2.75E-05	-0.0025	-7.60E-04
4	u	1.26E-04	1.26E-04	1.40E-04	8.69E-05	3	ϵ_x			0.0057	1.95E-03
	v	-2.38E-05	-2.38E-05	-4.14E-05	-3.90E-05		ϵ_y			-0.0012	-5.87E-04
5	u	7.37E-05	6.95E-05	5.43E-05	4.78E-05		γ_{xy}			0.0003	1.69E-04
	v	-1.21E-05	-2.74E-05	-2.79E-05	-4.41E-05	4	ϵ_x			0.0061	1.29E-03
6	u	1.27E-04	1.26E-04	4.78E-05	8.93E-05		ϵ_y			-0.0017	-7.14E-04
	v	-1.62E-05	-1.11E-05	-4.41E-05	-3.17E-05		γ_{xy}			-0.0003	-3.69E-03
7	u	6.95E-05	1.27E-04	8.93E-05	9.30E-05						
	v	-2.74E-05	-1.62E-05	-3.17E-05	-3.24E-05						
8	u	4.01E-05	7.37E-05	8.69E-05	5.43E-05						
	v	-2.34E-05	-1.21E-05	-3.90E-05	-2.79E-05						
9	u	7.78E-05	7.78E-05	1.14E-04	7.30E-05						
	v	-1.73E-05	-1.73E-05	-2.58E-05	-3.37E-05						
10	u			1.14E-04	1.40E-04						
	v			-4.14E-05	-4.14E-05						
11	u			6.07E-05	1.14E-04						
	v			-4.14E-05	-4.14E-05						
12	u			7.30E-05	1.40E-04						
	v			-3.36E-05	-2.52E-05						
13	u			9.30E-05	1.14E-04						
	v			-3.24E-05	-2.58E-05						
14	u			1.16E-04	1.41E-04						
	v			-3.29E-05	-3.29E-05						
15	u			1.41E-04	1.16E-04						
	v			-3.29E-05	-3.29E-05						

Table 2. comparison between Matlab and ANSYS result of displacement and strain obtained for the 6-noded; 2 and 4 elements

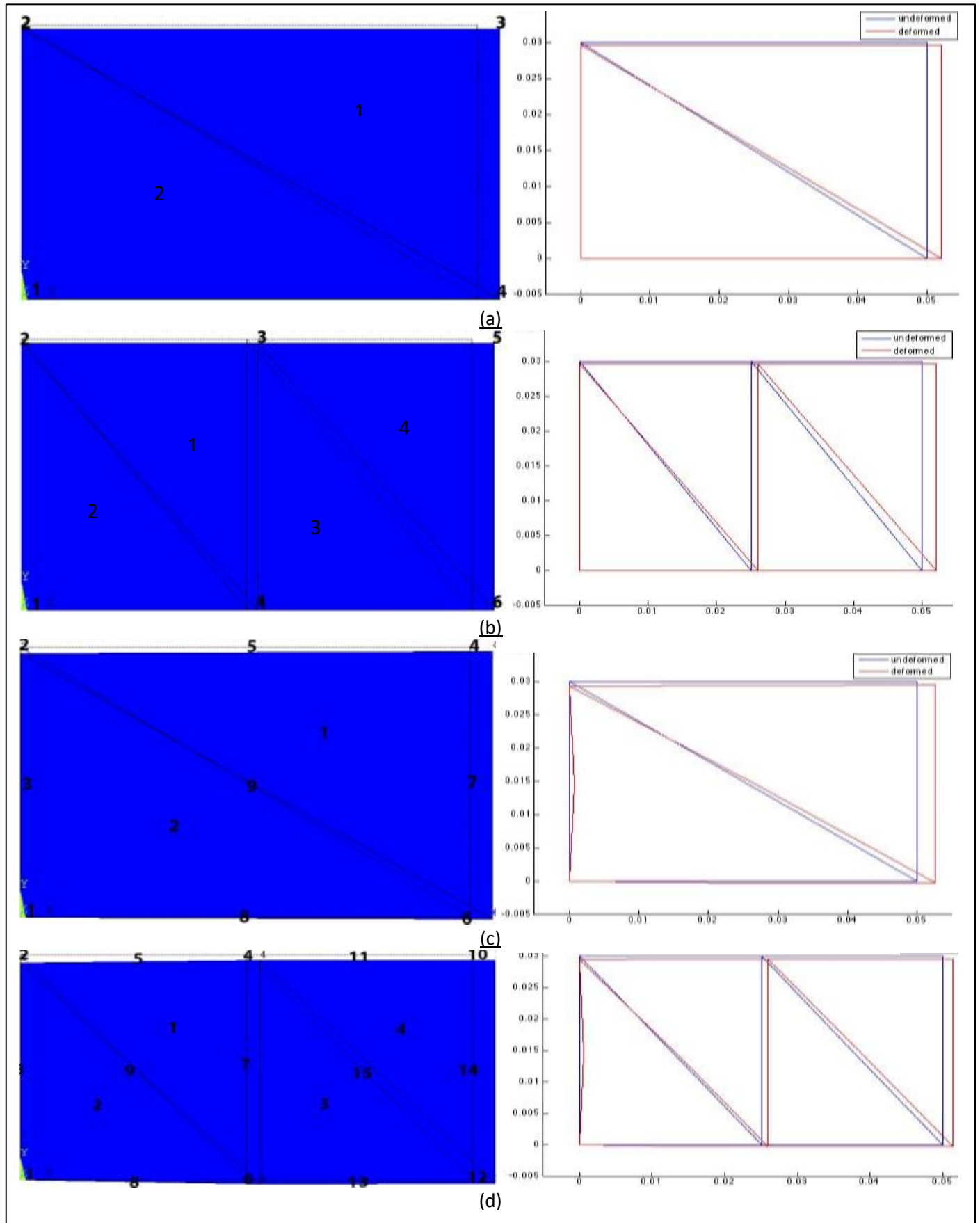


Figure 4. (a) shows the deformation shape of the plate from ANSYS and Matlab for the 3-noded; 2 elements and (b) for the 4 elements. while (c) and (d) show the deformation in the 6-noded; 2 elements and 4 elements respectively.

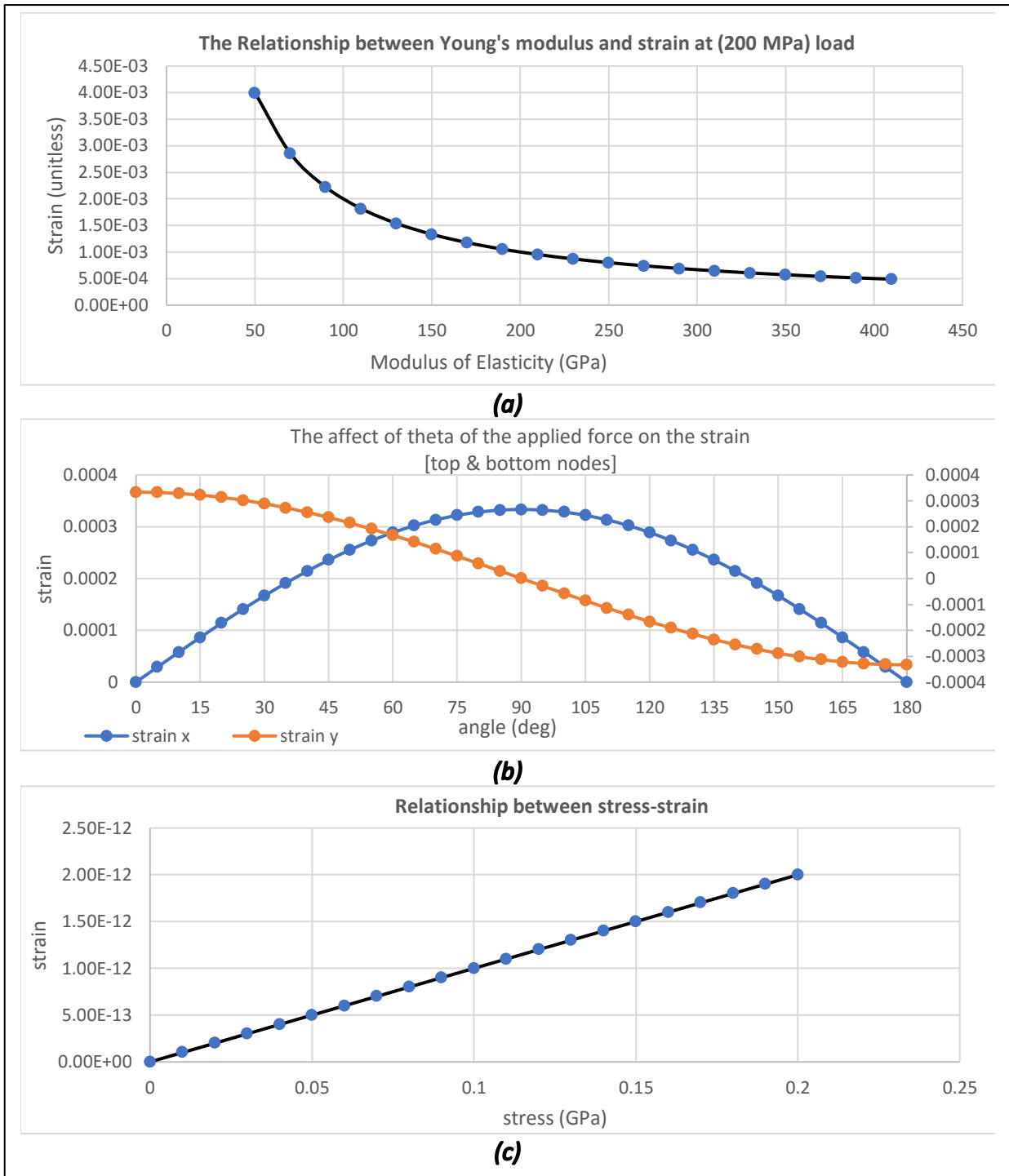


Figure 5. (a) represents the relationship between the elastic modulus and strain based on the dimension given. And (b) shows the impact of the orientation of the load applied on the strain. And (c) represents the stress-strain relationship (graph).

Appendix

%%% PART 2 %%(i) 4-ELEMENTS - 3-NODED

```
d c;
E = 100e9;          %YOUNG S MODULUS
v = 0.3;            %POISSON S RATIO
t = 0.002;          %THICKNESS ( m)
P = 200e6;           %PRESSURE APPLIED ( Pa)
A = 0.5*0.025*0.03; %AREA ( m^2)
f =( P*t*0.03)/2;    %FORCE APPLIED ( N)
D =( E/(1-(v^2)))*[1, v, 0; v, 1, 0; 0, 0, ((1-
v)/2)];              %STRESS- STRAIN
                        %RELATIONSHIP
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%ELEMENT 1
x11=0;
y11=0;
x21=0.025;
y21=0;
x31=0;
y31=0.03;
```

%%SHAPE FUNCTION COEFFICIENTS

```
ai 1 =(x21*y31) - (x31*y21);
aj 1 =(x31*y11) - (x11*y31);
ak1 =(x11*y21) - (x21*y11);
```

```
bi 1 = y21- y31;
bj 1 = y31- y11;
bk1 = y11- y21;
```

```
ci 1 = x31- x21;
cj 1 = x11- x31;
ck1 = x21- x11;
```

%%JACOBIAN MATRIX

```
j1 =[cj 1, bj 1, ci 1, bi 1];
J1 = det(j 1);
```

%%STRAIN-DISPLACEMENT RELATION & STIFFNESS MATRIX

```
B1 =(1/J1)*[bi 1, 0, bj 1, 0, bk1, 0, 0, ci 1, 0, cj 1, 0, ck1;...
            d1, bi 1, cj 1, bj 1, ck1, bk1]
```

```
k1 =t*A*B1'*D*B1
k1_global = zeros (12,12);
k1_global ([1,2,3,4,9,10],[1,2,3,4,9,10]) = ...
            k1([1,2,5,6,3,4],[1,2,5,6,3,4])
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

%ELEMENT 2

```
x12=0.025;
y12=0;
x22=0.025;
y22=0.03;
x32=0;
y32=0.03;
```

%%SHAPE FUNCTION COEFFICIENTS

```
ai 1 =(x22*y32) - (x32*y22);
aj 1 =(x32*y12) - (x12*y32);
ak1 =(x12*y22) - (x22*y12);
```

```
bi 2 = y22- y32;
bj 2 = y32- y12;
bk2 = y12- y22;
```

```
ci 2 = x32- x22;
cj 2 = x12- x32;
ck2 = x22- x12;
```

%%JACOBIAN MATRIX

```
j2 =[cj 2, bj 2, ci 2, bi 2];
J2 = det(j 2);
```

%%STRAIN-DISPLACEMENT RELATION & STIFFNESS MATRIX

```
B2 =(1/J2)*[bi 2, 0, bj 2, 0, bk2, 0, 0, ci 2, 0, cj 2, 0, ck2;...
            d2, bi 2, cj 2, bj 2, ck2, bk2]
```

```
k2 =t*A*B2'*D*B2
k2_global = zeros (12,12);
k2_global ([3,4,9,10,11,12],[3,4,9,10,11,12]) = ...
            k2 ([5,6,1,2,3,4],[5,6,1,2,3,4])
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%ELEMENT 3
x13=0.025;
y13=0;
x23=0.05;
y23=0;
x33=0.025;
y33=0.03;
```

%%SHAPE FUNCTION COEFFICIENTS

```
ai 3 =(x23*y33) - (x33*y23);
aj 3 =(x33*y13) - (x13*y33);
ak3 =(x13*y23) - (x23*y13);
```

```
bi 3 = y23- y33;
bj 3 = y33- y13;
bk3 = y13- y23;
```

```
ci 3 = x33- x23;
cj 3 = x13- x33;
ck3 = x23- x13;
```

%%JACOBIAN MATRIX

```
j3 =[cj 3, bj 3, ci 3, bi 3];
J3 = det(j 3);
```

%%STRAIN-DISPLACEMENT RELATION & STIFFNESS MATRIX

```
B3 =(1/J3)*[bi 3, 0, bj 3, 0, bk3, 0, 0, ci 3, 0, cj 3, 0, ck3;...
            d3, bi 3, cj 3, bj 3, ck3, bk3]
```

```
k3 =t*A*B3'*D*B3
k3_global = zeros (12,12);
k3_global ([5,6,9,10,11,12],[5,6,9,10,11,12]) = ...
            k3 ([3,4,1,2,5,6],[3,4,1,2,5,6])
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

%ELEMENT 4

```
x14=0.05;
y14=0;
x24=0.05;
y24=0.03;
x34=0.025;
y34=0.03;
```

%%SHAPE FUNCTION COEFFICIENTS

```
ai 4 =(x24*y34) - (x34*y24);
aj 4 =(x34*y14) - (x14*y34);
ak4 =(x14*y24) - (x24*y14);
```

```
bi 4 = y24- y34;
bj 4 = y34- y14;
bk4 = y14- y24;
```

```
ci 4 = x34- x24;
cj 4 = x14- x34;
ck4 = x24- x14;
```

%%JACOBIAN MATRIX

```
j4 =[cj 4, bj 4, ci 4, bi 4];
J4 = det(j 4);
```

%%STRAIN-DISPLACEMENT RELATION & STIFFNESS MATRIX

```
B4 =(1/J4)*[bi 4, 0, bj 4, 0, bk4, 0, 0, ci 4, 0, cj 4, 0, ck4;...
            d4, bi 4, cj 4, bj 4, ck4, bk4]
```

```
k4 =t*A*B4'*D*B4
k4_global = zeros (12,12);
k4_global ([5,6,7,8,11,12],[5,6,7,8,11,12]) = ...
            k4 ([1,2,3,4,5,6],[1,2,3,4,5,6])
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

%SUM OF GLOBAL [K]

```
K = k1_global + k2_global + k3_global + k4_global
```

```

%%APPLY BOUNDARY CONDITIONS - [u1=v1=u2=0]
disp(' u1=v1=u2=0 (due to constraints) THEREFORE:')
K_e = K(4:12, 4:12)

%%DISPLACEMENT [U] & STRAIN[S] CALCULATION - [KQ=F]
F=[Q; f; Q; f; Q; Q; Q; Q; Q; Q; Q; Q]; %%FORCE MATRIX
U=inv(K_e)*F; %%DISPLACEMENT
u1=Q;
v1=Q;
u2=Q;
v2=U(1,1);
u3=U(2,1);
v3=U(3,1);
u4=U(4,1);
v4=U(5,1);
u5=U(6,1);
v5=U(7,1);
u6=U(8,1);
v6=U(9,1);

disp(' displacement');
disp(' x y');
U_=[u1 v1; u2 v2; u3 v3; u4 v4; u5 v5; u6 v6]
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
disp(' ELEMENT 1 : DISPLACEMENT & FORCES')
U_1=[u1; v1; u3; v3; u2; v2]
f1=k1*U_1
disp(' ELEMENT 2 : DISPLACEMENT & FORCES')
U_2=[u3; v3; u4; v4; u2; v2]
f2=k2*U_2
disp(' ELEMENT 3 : DISPLACEMENT & FORCES')
U_3=[u3; v3; u5; v5; u4; v4]
f3=k3*U_3
disp(' ELEMENT 4 : DISPLACEMENT & FORCES')
U_4=[u5; v5; u6; v6; u4; v4]
f4=k4*U_4
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
disp(' ELEMENT 1 : STRAIN')
S1=B1*U_1
disp(' ELEMENT 2 : STRAIN')
S2=B2*U_2
disp(' ELEMENT 3 : STRAIN')
S3=B3*U_3
disp(' ELEMENT 4 : STRAIN')
S4=B4*U_4
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%PART 2 %%4 elements - 6 NODED
dc;
E=100*10^9; %%YOUNG'S MODULUS
v=0.3; %%POISSON'S RATIO
t=0.002; %%THICKNESS (m)
A=0.5*0.025*0.03; %%AREA (m^2)
P=200*10^6; %%PRESSURE APPLIED (Pa)
f=(P*t*0.03); %%FORCE APPLIED (N)
H=1/6; %%WEIGHTING FUNCTION
D=(E/(1-(v^2)))*[1, v, 0; v, 1, 0; 0, 0, ((1-v)/2)]; %%STRESS-STRAIN
%%RELATIONSHIP
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%ELEMENT 1
x11=Q;
y11=Q;
x21=0.025;
y21=Q;
x31=Q;
y31=0.03;
x41=0.0125;
y41=Q;
x51=0.0125;
y51=0.015;
x61=Q;
y61=0.015;
%%COORDINATES MATRIX E1

```

```

xi_yi1=[x11 y11; x21 y21; x31 y31; x41 y41; x51 y51; x61 y61];

```

```

%%L1 and L2 VALUES
L1=0.5;
L2=0.5;

```

```

%%STRAIN DISPLACEMENT RELATION & STIFFNESS MATRIX X 1
B=element1b_6N(xi_yi1, L1, L2);
B11=B;
k=element1k_6N(xi_yi1, H, t, D, L1, L2);
k11=k;

```

```

%%L1 and L2 VALUES
L1=0.5;
L2=Q;
%%STRAIN DISPLACEMENT RELATION & STIFFNESS MATRIX X 2
B=element1b_6N(xi_yi1, L1, L2);
B21=B;
k=element1k_6N(xi_yi1, H, t, D, L1, L2);
k21=k;

```

```

%%L1 and L2 VALUES
L1=Q;
L2=0.5;
%%STRAIN DISPLACEMENT RELATION & STIFFNESS MATRIX X 3
B=element1b_6N(xi_yi1, L1, L2);
B31=B;
k=element1k_6N(xi_yi1, H, t, D, L1, L2);
k31=k;

```

```

B1=B11+B21+B31; %%B MATRIX ELEMENT 1
K1=k11+k21+k31; %%LOCAL [K] ELEMENT 1
%%LOCAL TO GLOBAL [K] MATRIX 1
K1_global=zeros(30);
K1_global([1,2,3,4,9,10,13,14,21,22,23,24],[1,2,3,4,9,10,13,14,21,22,23,24])=...
K1([1,2,5,6,7,8,3,4,11,12,9,10],
[1,2,5,6,7,8,3,4,11,12,9,10]);
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%ELEMENT 2
x12=0.025;
y12=0.03;
x22=Q;
y22=0.03;
x32=0.025;
y32=Q;
x42=0.0125;
y42=0.03;
x52=0.0125;
y52=0.015;
x62=0.025;
y62=0.015;
%%COORDINATES MATRIX E2
xi_yi2=[x12 y12; x22 y22; x32 y32; x42 y42; x52 y52; x62 y62];

```

```

%%L1 and L2 VALUES
L1=0.5;
L2=0.5;
%%STRAIN DISPLACEMENT RELATION & STIFFNESS MATRIX X 1
B=element2b_6N(xi_yi2, L1, L2);
B12=B;
k=element2k_6N(xi_yi2, H, t, D, L1, L2);
k12=k;

```

```

%%L1 and L2 VALUES
L1=0.5;
L2=Q;
%%STRAIN DISPLACEMENT RELATION & STIFFNESS MATRIX X 2
B=element2b_6N(xi_yi2, L1, L2);
B22=B;
k=element2k_6N(xi_yi2, H, t, D, L1, L2);

```

[illegible]

